

Nikhil Singh

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SKILLS

Languages: Rust, Nix (Flakes), Python (uv), Go, Haskell, C/C++, Bash

Systems & Infra: Linux (NixOS), Kubernetes, Docker, AWS, QEMU, CI/CD

Developer Tools: Git, Neovim/Nixvim, Tmux, QEMU, SSH

Datastores & Messaging: ClickHouse, Kafka, Redis/Valkey, MySQL, MongoDB

Backend Frameworks: Actix Web (Rust), FastAPI, Node.js, Servant (Haskell)

Engineering Focus: Low-latency systems, reproducible infrastructure, distributed data pipelines

EXPERIENCE

Product Engineer | *Juspay Technologies*

Jun 2025 – Present

- Spearheaded the end-to-end architecture and development of **Project Bernoulli**, a mission-critical Rust payment gateway bridging PSPs and client SDKs, seamlessly processing 2M+ daily transactions with 99.99% uptime.
- Engineered Bernoulli's high-throughput event pipelines using **Kafka** and **ClickHouse**, optimizing SDK session management with sub-millisecond data caching via **Redis/Valkey**.
- Developed the backend processing pipeline for **UDIR** (Unified Dispute and Issue Resolution) refunds, automating dispute handling and significantly accelerating merchant refund lifecycles.
- Spearheaded the Nixification of the **Hyperswitch** payment router to ensure reproducible builds, and deployed a company-wide Nix devbox slashing onboarding time for 800+ engineers from hours to minutes.
- Engineered the **juspay-mcp** service to orchestrate deterministic Python 3.13 environments via **uv2nix**, and deployed Nix-hosted fine-tuned LLMs, reducing manual PR reviews by 80%.
- Authored a custom GPG library in Rust with MDC payload support for secure cryptographic operations in payment flows.

M.Tech Researcher – Systems Engineering | *Gautam Buddha University*

May 2023 – May 2025

- Authored IEEE paper on modular NixOS configurations for reproducible OS management (IEEE Xplore).
- Co-authored IEEE paper on ML-based heavy rainfall prediction for the Northern Himalayan region.
- Developed Nix flake-based configurations using **disko** and **home-manager** for hardware-agnostic deployments.

Software Developer Intern | *Decet Technology*

2023 – 2024

- Slashed deployment times from 2+ hours to under 5 minutes by engineering a custom Go-based CI/CD infrastructure, directly accelerating the backend release cycle.
- Built core Python backend services handling 10k+ daily requests with REST APIs and database integrations.

Package Maintainer – NixOS (nixpkgs) & Arch Linux | *NixOS & Arch Communities*

2020 – Present

- Upstreamed a highly adopted module to **home-manager**, enabling declarative user environment configurations for a large segment of the Nix community.
- Maintain multiple open-source Nix flake projects and 15+ packages across **nixpkgs** and the AUR with 50k+ combined downloads; host and manage a custom binary cache.
- Contributed 10+ upstream fixes to core ecosystem projects including NixOS and Hyprwm.

PROJECTS

Ndots | *NixOS System Configuration (100+ Stars)*

2024 – Present

- Modular NixOS flake orchestrating reproducible configurations across multiple personal and organizational systems, supporting diverse machine architectures.
- Automated deployment pipelines integrating **sops** for secure secret management, reducing system setup time from days to under 30 minutes with full rollback capability.

Nvim | *Nixvim/Neovim Distribution for Terminal Users*

2024 – Present

- Maintained a high-performance Nixvim configuration integrated with custom highlight logic, optimized for keyboard-centric workflows.
- Modular plugin architecture supporting LSP, DAP, and Treesitter for 20+ languages with sub-50ms startup times.

Tmux Minimal Status | *Terminal UI, Open Source*

2020 – Present

- Maintainer of popular tmux status bar (240+ GitHub stars); optimized for minimal resource footprint in high-frequency terminal environments.

EDUCATION

Gautam Buddha University

Greater Noida, UP

Master of Technology in Software Engineering (**CGPA: 8.83**)

Aug 2023 – May 2025

Bachelor of Technology in Computer Science Engineering

Aug 2020 – Aug 2023